SRM INSTITUTE OF SCIENCE & TECHNOLOGY
Deemed to be University u/s 3 of UGC Act, 1956

21CSC305P
MACHINE LEARNING

Grouping Warehouse Products with K-Means Clustering

Project Review 3 | Unit 3

Team: Cash guard (No. 17)

V. Anbu Chelvan, S Roopesh, B. Charith Reddy, Y Rupesh Reddy

13 October 2026

Our problem statement

Group warehouse products (SKUs) that behave alike, so managers can place them smartly and workers walk less while picking orders.

We group the SKUs using four things

1

Pick frequency

How often it is picked for orders

2

Seasonal demand

How fast it sells in each season

3

Physical size

Length, width, height and weight

4

Handling needs

Fragile, heavy or needs special care

Goal: better inventory placement and less travel time during order fulfilment

Key terms in simple words

SKU

Stock Keeping Unit: one unique product type stored in the warehouse.

Clustering

Dividing data into groups where items in a group are very similar.

Pick frequency

How many times a product is taken from the shelf to fill orders.

Centroid

The centre (mean point) of a cluster. Each cluster has one.

Seasonal demand velocity

How fast a product sells in a season. Example: umbrellas in the monsoon.

Unsupervised learning

Learning from data that has no labels. The algorithm finds the groups itself.

Unit 3 solutions we reviewed

Centroid-based

K-Means clustering

Splits data into k groups. Each item joins the nearest centre.

Our choice

Distribution-based

Mixtures of Gaussians (EM)

Soft clustering. Each item gets a probability for every group.

Tree-based

Hierarchical clustering

Builds a tree of clusters by merging or splitting step by step.

Fewer features

PCA and Factor analysis

Reduces many features to a few. It does not make groups.

From the Unit 3 syllabus: K-means, mixtures of Gaussians, EM, factor analysis, PCA, hierarchical clustering

Our choice: K-Means clustering

DEFINITION

K-Means groups similar data points into k clusters without needing labelled data.

k = the number of groups we want



Why it suits our project

1. Our task says "partition". K-Means is a partitioning method.
2. Every SKU gets exactly one group, just like one storage zone.
3. All four inputs can be written as numbers, so distance works.
4. Simple, fast for thousands of SKUs, easy to re-run each season.

Why not the other solutions

Method	What it gives	Fit for our problem
Mixtures of Gaussians (EM)	A probability for every group (soft)	A product cannot sit 60% in one zone and 40% in another.
Hierarchical clustering	A tree of clusters (dendrogram)	Compares every pair of items, so it gets slow for thousands of SKUs.
PCA / Factor analysis	Fewer features	Does not create groups. It can only help before clustering.
K-Means clustering	**k clear groups, one centre each**	**Fits our need: one clear group for every SKU.**

How K-Means works

1 → 2 → 3 → 4 → 5

1 Pick k centres

Choose k centroids at random to start.

2 Measure distance

Find the distance from each SKU to every centroid.

3 Assign

Put each SKU in the cluster with the nearest centroid.

4 Update centres

Move each centroid to the mean of its SKUs.

5 Repeat

Do steps 2 to 4 again until no SKU changes its cluster.

Distance used: Euclidean distance, the straight-line distance between two points.

Step 1: Prepare the SKU data

Real data: Olist public e-commerce dataset, 99,441 orders (2016 to 2018)

SKU	Orders	Season index	Volume (L)	Weight (kg)
aca2eb7d	527	2.8	15.0	2.6
42a2c92a	183	4.8	33.0	7.0
54d9ac71	114	8.0	7.8	1.9
dbb67791	88	4.5	2.7	0.2

Season index = share of orders in the busiest month x 12

Source: **github.com/olist/work-at-olist-data** (Olist, MIT licence)

1. **383 real SKUs**
   Products ordered at least 30 times, out of 32,951.

2. **Four numbers each**
   Orders, season index, volume, and weight for handling.

3. **Scale the features**
   Log of the uneven columns, then all on the same scale.

The real data at a glance

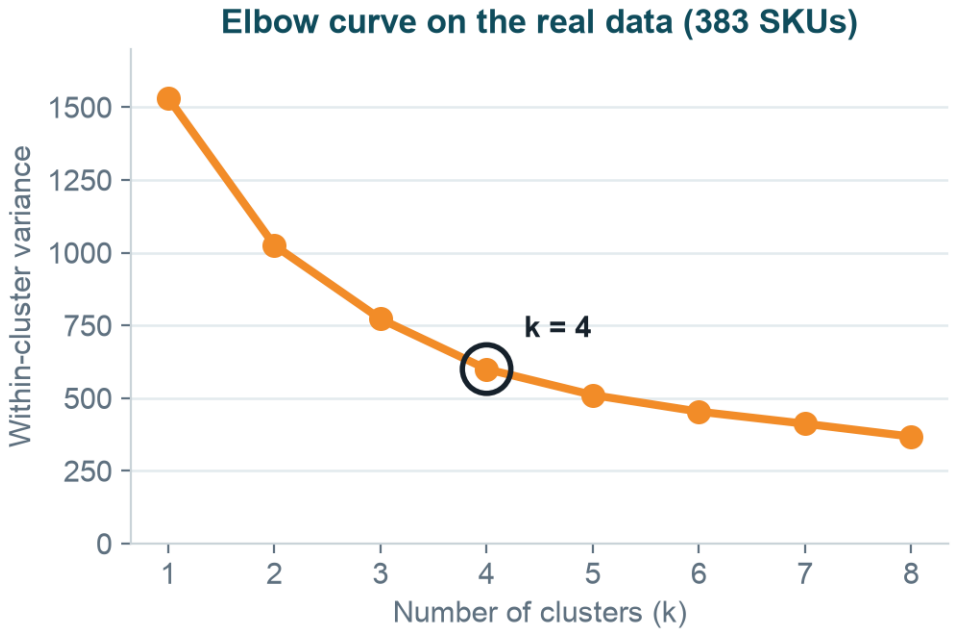
99,441
real orders, 2016 to 2018

32,951
products in the dataset

383
SKUs ordered 30 times or more



Step 2: Choose the value of k



1 Try many values

Run K-Means for k = 1 to 8. Note the error each time.

2 Plot the curve

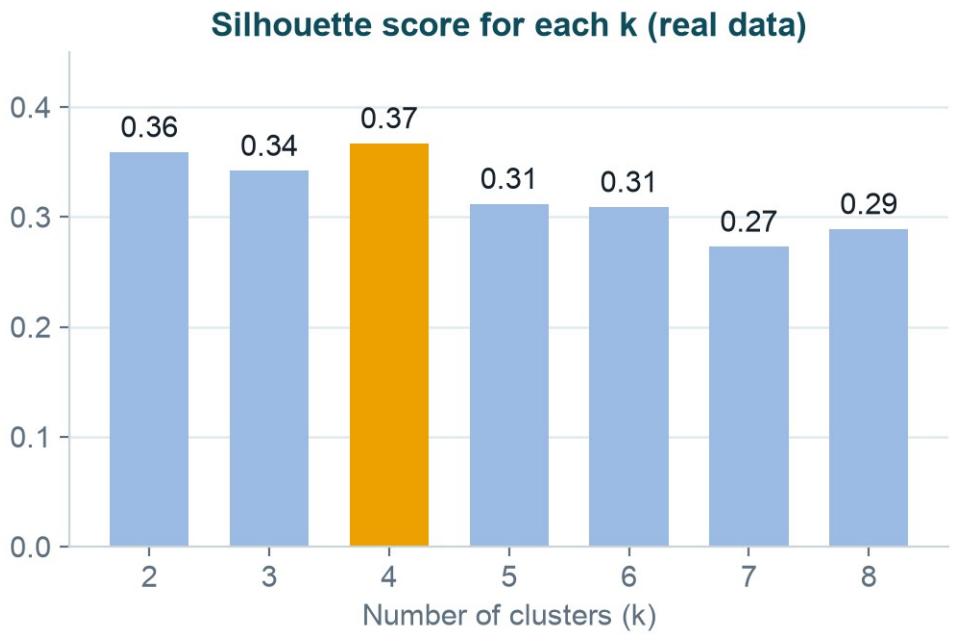
Error = within-cluster variance. It falls as k grows.

3 Find the elbow

The error falls 22% at k = 4, then only 15% at k = 5.

On the real data: k = 4

A second check: silhouette



1 What it measures

How well each SKU sits inside its own group. Higher is better.

2 Highest at k = 4

The score is 0.37 at k = 4, above every other k.

3 Both checks agree

The elbow and the silhouette both point to 4 clusters.

Both checks: k = 4

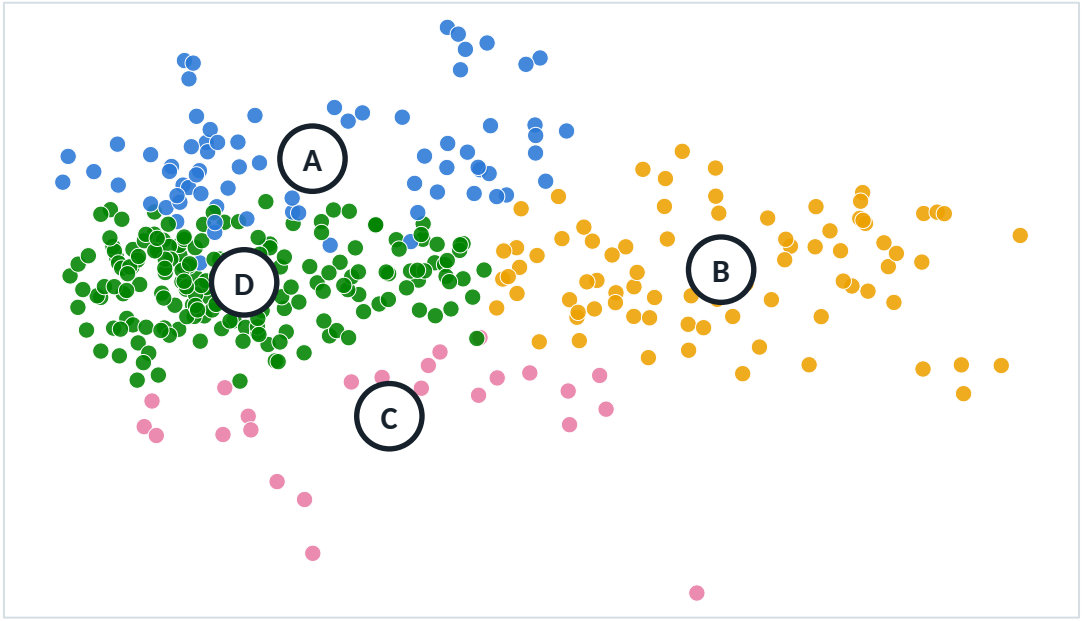
Step 3: Clusters become zones

Cluster	SKUs in it	Where to place
A Fast movers	76 SKUs, about 160 orders each	Front racks, near packing
B Bulky / heavy	84 SKUs, about 47 L and 7.6 kg	Ground level, low racks
C Seasonal	25 SKUs, season index 7.4	Flexible zone, moved forward in season
D Regular small	198 SKUs, about 45 orders each	Back racks

Real clusters: K-Means with k = 4 on 383 Olist SKUs.



The four clusters in one picture



1 Each dot is one SKU

383 real products, coloured by their cluster.

2 Drawn with PCA

Four numbers squeezed into two. It keeps 76% of the variation.

3 Same colour, same place

Similar products landed in the same cluster.

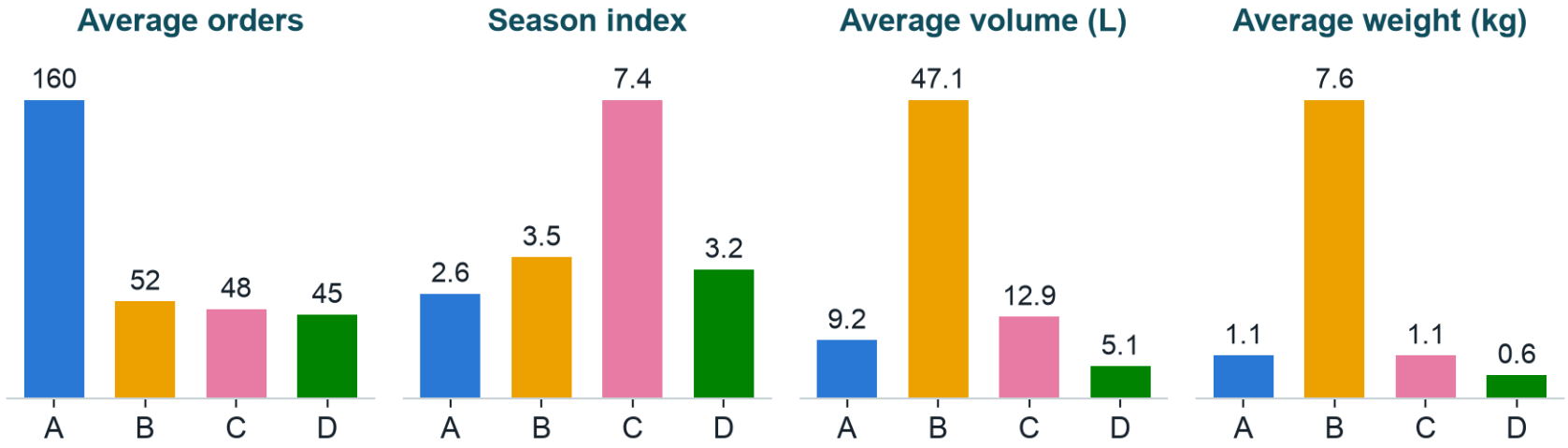
A Fast movers

B Bulky / heavy

C Seasonal

D Regular small items

What each cluster looks like



A Fast movers

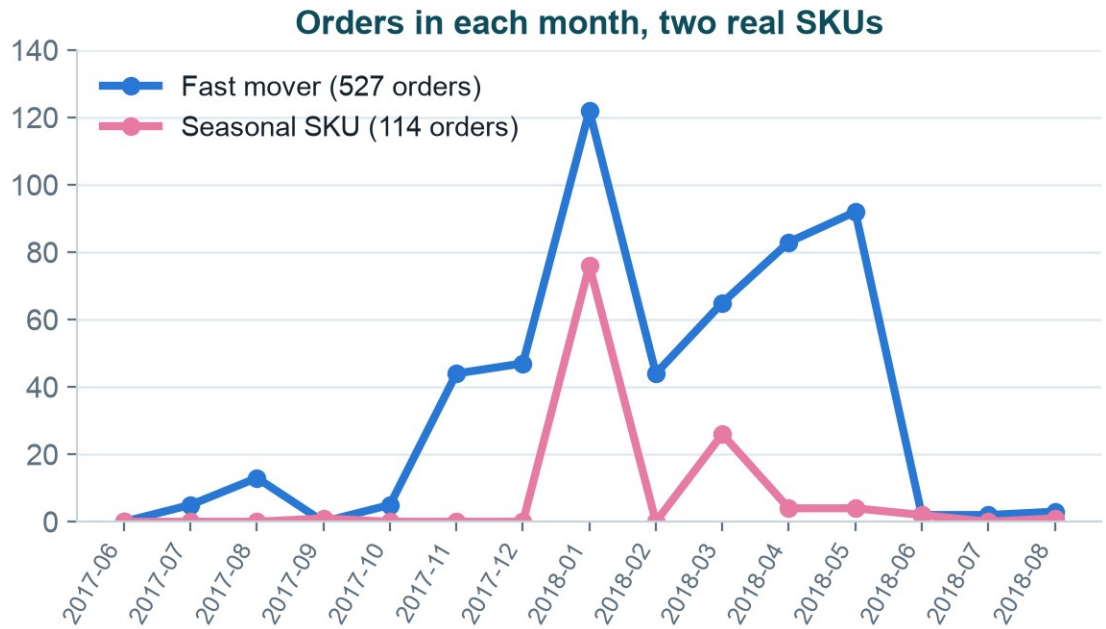
B Bulky / heavy

C Seasonal

D Regular small items

A is picked the most. B is the biggest and heaviest. C sells in one season. D is small and steady.

Seasonal against fast mover



1 **Seasonal SKU**

76 of its 114 orders came in a single month, January 2018.

2 **Fast mover**

527 orders, with 40 or more in seven different months.

3 **Why it matters**

A seasonal product needs a front place only in its season.

How this solves our problem

1

Less walking

Fast movers sit closest to packing. In our simple model, walk per pick falls by 27%.

2

Better placement

Bulky and heavy products get their own ground-level zone.

3

Season ready

Seasonal products are one group, easy to move forward in season.

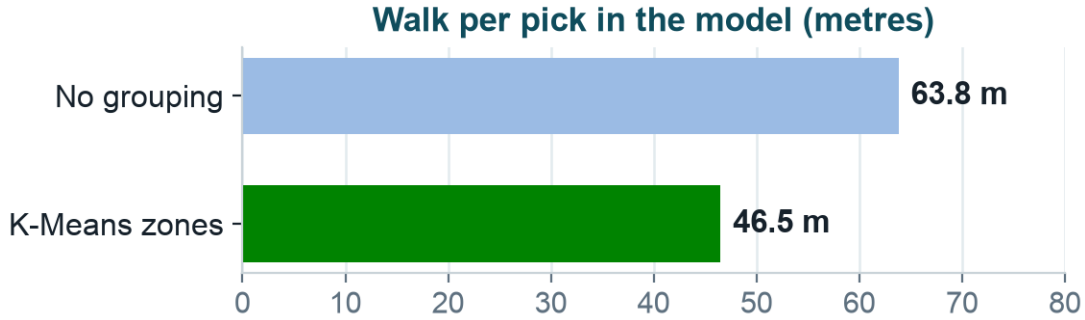
How we check the clusters: each centroid should read like a typical product. Fast movers: about 160 orders, 9 litres, 1 kg. The walking figure is a model, not a measurement.

The aisle and the walk per pick

One aisle of 128 m, from the packing desk



A Fast movers B Bulky / heavy C Seasonal D Regular small items



27%

less walking per pick

A simple model, not a measurement

Conclusion

1. Problem: group warehouse SKUs so that pickers travel less.
2. Chosen from Unit 3: K-Means clustering, a simple partitioning method.
3. Tested on real data: 383 SKUs gave 4 clear clusters that map to zones.

Live demo and Colab code:
zanyanbu.github.io/warehouse-sku-kmeans

Thank you

Questions are welcome

